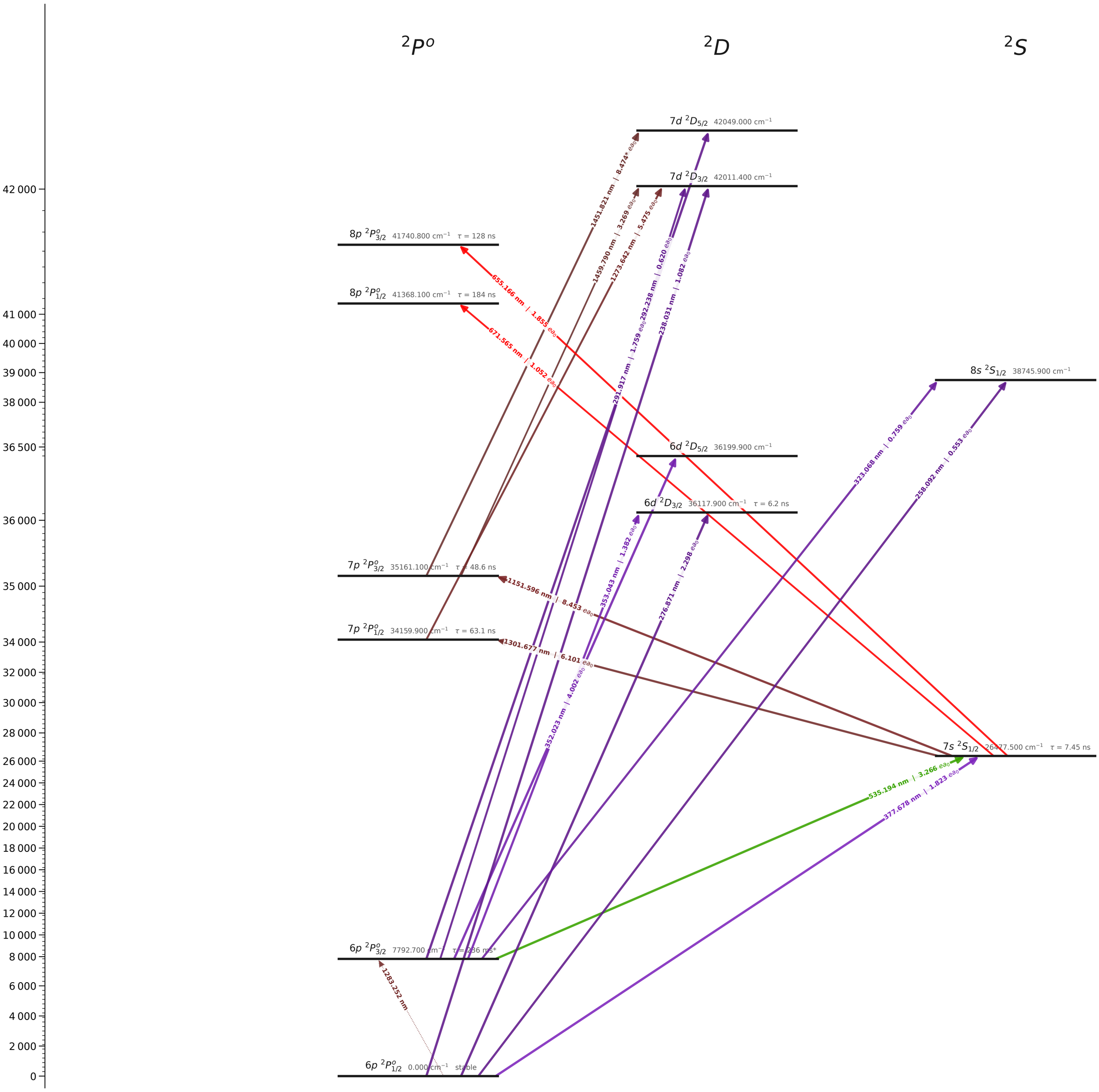
Thallium

Tl (Tl I). Strongest classified transitions below 2 μm (NIST): 12 levels, 18 lines.

Tl^+ ionisation limit 49266.66 cm^{-1} = 6.10829 eV (202.977 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| er \| J' \rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient *A*. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (wh



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\lvert\langle J\|\|er\|\|J'\rangle\rvert$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$6p\ ^2P^o_{3/2}$ – $6d\ ^2D_{5/2}$	352.0234	351.9228	851.6264	4.002	1.240e+08	–	–	352 nm proposed laser-cooling transition 6P3/2
$6p\ ^2P^o_{1/2}$ – $7s\ ^2S_{1/2}$	377.6780	377.5707	793.777941	1.823	6.250e+07	21.36	46.3	377.6 nm resonance line
$6p\ ^2P^o_{3/2}$ – $7s\ ^2S_{1/2}$	535.1944	535.0456	560.1562	3.266	7.050e+07	21.36	53.7	535.0 nm green line
$7s\ ^2S_{1/2}$ – $8p\ ^2P^o_{3/2}$	655.1663	654.9854	457.5822	1.855	6.200e+06	1.246	79.2	655 nm second step of two-step spectroscopy
$7s\ ^2S_{1/2}$ – $8p\ ^2P^o_{1/2}$	671.5646	671.3793	446.4090	1.052	3.700e+06	0.8645	68.1	671 nm second step of two-step spectroscopy
$6p\ ^2P^o_{1/2}$ – $6p\ ^2P^o_{3/2}$	1283.2523	1282.9013	233.6193	–	4.094e+00	6.735e-07	96.7	1283 nm M1
$7s\ ^2S_{1/2}$ – $7p\ ^2P^o_{1/2}$	1301.6766	1301.3206	230.3126	6.101	1.710e+07	2.522	100	1301 nm second step of two-step

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{205}Tl hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$6p\ ^2P^o_{1/2}$	21310.835(5)	–	F=0: -15983.13 F=1: +5327.71
$6p\ ^2P^o_{3/2}$	265.03830(10)	–	F=1: -331.30 F=2: +198.78
$7s\ ^2S_{1/2}$	12296.1(7)	–	F=0: -9222.07 F=1: +3074.02
$7p\ ^2P^o_{1/2}$	2173.3(8)	–	F=0: -1629.97 F=1: +543.33
$7p\ ^2P^o_{3/2}$	311.4(3)	–	F=1: -389.25 F=2: +233.55
$6d\ ^2D_{3/2}$	-42.9(4)	–	F=1: +53.62 F=2: -32.17
$6d\ ^2D_{5/2}$	226.9(3)	–	F=2: -397.07 F=3: +283.62
$8s\ ^2S_{1/2}$	3870.8(14)	–	F=0: -2903.10 F=1: +967.70
$8p\ ^2P^o_{1/2}$	788.5(6)	–	F=0: -591.38 F=1: +197.12
$8p\ ^2P^o_{3/2}$	131.69(20)	–	F=1: -164.61 F=2: +98.77
$7d\ ^2D_{3/2}$	-56.3(6)	–	F=1: +70.38 F=2: -42.23
$7d\ ^2D_{5/2}$	180.20(20)	–	F=2: -315.35 F=3: +225.25

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$6p\ ^2P^o_{1/2}$ – $7s\ ^2S_{1/2}$	377.678	205-203	1658.3(7)
$7s\ ^2S_{1/2}$ – $8p\ ^2P^o_{3/2}$	655.166	203-205	463.4(7)
$7s\ ^2S_{1/2}$ – $8p\ ^2P^o_{1/2}$	671.565	203-205	450.1(8)
$7s\ ^2S_{1/2}$ – $7p\ ^2P^o_{1/2}$	1301.677	203-205	534.4(9)

Shift = ν(first isotope) − ν(second isotope).